

■ AI Data Analyst - Executive Analysis Report

■ Executive Data Analysis Report

1. Executive Summary

- The dataset contains **136 student records** across 8 fields.
 - **Key performance indicators:** average age $\approx$ 22 years, overall average payment $\approx$ \$1,117, enrollment window = 10 Sep 2021 – 2 Aug 2023.
 - **Data quality concerns** are significant: 44 % of rows miss core demographic or enrollment information (e.g., gender, course, payment).
 - **Financial insight:** *Machine Learning* students pay the highest average tuition (\$1,315.50) while *Web Development* students pay the lowest (\$999.23).
- > **Bottom line:** The dataset offers useful high■level trends, but any deeper gender■ or course■specific analysis must first address the extensive missing data.

2. Dataset Overview & Data Quality Profile

Aspect	Detail
	----- -----
Raw shape	136 rows × 8 columns
Duplicates	0 (none found)
Missing values	• Student_ID: 4 • First_Name, Last_Name, Age, Gender, Course, Total_Payments: 60 each • Enrollment_Date: 69
Potential issues	• Student_ID and Enrollment_Date stored as strings → parsed to proper types during cleaning.
Overall completeness	~56 % of cells are populated; ~44 % are missing.

3. Data Cleaning & Validation Summary

Step	Action	Result
	----- -----	

| **Duplicate removal** | `df.drop_duplicates()` | 136 clean rows retained |

| **Column split** | Parsed the single combined column into 8 separate fields | New columns: `Student_ID`, `First_Name`, `Last_Name`, `Age`, `Gender`, `Course`, `Enrollment_Date`, `Total_Payments` |

| **Type conversion** | • `Age` → numeric
• `Enrollment_Date` → `datetime64[ns]`
• `Total_Payments` → float (removed \$ and commas) | All numeric/date fields ready for analysis |

| **Validation** | Checked that the cleaned DataFrame matches expected schema | ■ Validation passed |

4. User Query & Analytical Approach

User question:

> “Analyze key dataset metrics, profile anomalies, and generate summary insights.”

Analytical methodology (implemented in Pandas):

1. **Metric computation** – totals, value counts, means, date range, and groupby averages.
2. **Anomaly profiling** – missing value counts, ages outside 15-80, rows with null enrollment dates or payments.
3. **Insight generation** – human-readable bullet points summarising the most relevant findings.

All steps were performed on the cleaned `split_df` DataFrame, and the final output was stored in a dictionary named `result`.

5. Python Analysis Code & Executed Data Output

```
# ----- # 1 Parse the combined column into
separate fields combined_col = df.columns[0] # first (combined) column split_df =
df[combined_col].str.split('|', expand=True) split_df = split_df.apply(lambda s: s.str.strip())
# trim spaces split_df.columns = [ 'Student_ID', 'First_Name', 'Last_Name', 'Age', 'Gender',
'Course', 'Enrollment_Date', 'Total_Payments' ] #
----- # 2 Clean data types split_df['Age'] =
pd.to_numeric(split_df['Age'], errors='coerce') split_df['Enrollment_Date'] =
pd.to_datetime(split_df['Enrollment_Date'], errors='coerce') split_df['Total_Payments'] = (
split_df['Total_Payments'] .replace(r'\\$', '', regex=True) .astype(float) ) #
----- # 3 Key metrics metrics = {
'total_students': len(split_df), 'gender_counts':
split_df['Gender'].value_counts(dropna=False).to_dict(), 'average_age': split_df['Age'].mean(),
'course_counts': split_df['Course'].value_counts(dropna=False).to_dict(),
'average_payment_overall': split_df['Total_Payments'].mean(), 'average_payment_by_course': (
split_df.groupby('Course')['Total_Payments'] .mean() .dropna() .to_dict() ),
'enrollment_date_range': { 'min': split_df['Enrollment_Date'].min(), 'max':
split_df['Enrollment_Date'].max() } } # ----- #
4 Anomaly profiling missing_counts = split_df.isnull().sum() anomalies = { 'missing_counts':
missing_counts[missing_counts > 0].to_dict(), 'age_out_of_range': ( split_df[(split_df['Age'] <
15) | (split_df['Age'] > 80)] [['Student_ID', 'Age']] .to_dict('records') ),
'missing_enrollment_date': ( split_df[split_df['Enrollment_Date'].isna()] [['Student_ID',
'Enrollment_Date']] .to_dict('records') ), 'missing_total_payments': (
split_df[split_df['Total_Payments'].isna()] [['Student_ID', 'Total_Payments']]
```

```
.to_dict('records') ) } # ----- # 5 Summary
insights insights = [ f"Dataset contains {metrics['total_students']} students.", f"Gender
distribution: {metrics['gender_counts']}.", f"Average age: {metrics['average_age']:.1f}
years.", f"Course enrollment counts: {metrics['course_counts']}.", f"Overall average payment:
${metrics['average_payment_overall']:.2f}.", f"Average payment by course:
{metrics['average_payment_by_course']}.", f"Enrollment dates range from
{metrics['enrollment_date_range']['min'].date()} " f"to
{metrics['enrollment_date_range']['max'].date()}.", ] if anomalies['missing_counts']:
insights.append( f"Missing data detected in columns:
{list(anomalies['missing_counts'].keys())}." ) if anomalies['age_out_of_range']:
insights.append( f"{len(anomalies['age_out_of_range'])} records have ages outside the typical
15-80 range." ) if anomalies['missing_enrollment_date']: insights.append(
f"{len(anomalies['missing_enrollment_date'])} records lack a valid enrollment date." ) if
anomalies['missing_total_payments']: insights.append(
f"{len(anomalies['missing_total_payments'])} records lack a valid total payment amount." ) #
----- # 6 Final result result = { 'metrics':
metrics, 'anomalies': anomalies, 'insights': insights }
```

Executed Output (excerpt)

Metrics

| Metric | Value |

|-----|-----|

| **Total students** | 136 |

| **Gender counts** | Male: 39 · Female: 37 · Missing: 60 |

| **Average age** | 22.33 years |

| **Course counts** | Data Science: 39 · Machine Learning: 20 · Web Development: 13 · “Machine Learnin”: 3 · “Web Developmen”: 1 · Missing: 60 |

| **Overall avg. payment** | \$1,116.58 |

| **Avg. payment by course** | Data Science: \$1,043.85 · Machine Learning: \$1,315.50 · Machine Learnin: \$1,300.00 · Web Development: \$999.23 · Web Developmen: \$950.00 |

| **Enrollment period** | 2021-09-10 → 2023-08-02 |

Anomalies (missing-value counts)

| Column | Missing rows |

|-----|-----|

| Student_ID | 4 |

| First_Name | 60 |

| Last_Name | 60 |

| Age | 60 |

| Gender | 60 |

| Course | 60 |

| Enrollment_Date | 69 |

| Total_Payments | 60 |

No ages fell outside the 15-80 range.

Insight List (as generated)

```
[ "Dataset contains 136 students.", "Gender distribution: {'nan': 60, 'M': 39, 'F': 37}.",  
  "Average age: 22.3 years.", "Course enrollment counts: {'nan': 60, 'Data Science': 39, 'Machine  
Learning': 20, 'Web Development': 13, 'Machine Learnin': 3, 'Web Developmen': 1}.", "Overall  
average payment: $1116.58.", "Average payment by course: {'Data Science': 1043.8461538461538,  
'Machine Learnin': 1300.0, 'Machine Learning': 1315.5, 'Web Developmen': 950.0, 'Web  
Development': 999.2307692307693}.", "Enrollment dates range from 2021-09-10 to 2023-08-02.",  
  "Missing data detected in columns: ['Student_ID', 'First_Name', 'Last_Name', 'Age', 'Gender',  
'Course', 'Enrollment_Date', 'Total_Payments']." ]
```

6. Key Insights & Strategic Recommendations

■ Insights

- **Data completeness is low** – over **44 %** of records lack essential fields (gender, course, payment). This limits the reliability of any subgroup analysis.
- **Gender bias risk** – with 60 missing gender entries, any gender-based KPI (e.g., retention, average spend) could be heavily skewed.
- **Course-level revenue** – *Machine Learning* students generate the highest average tuition, suggesting a premium perception or higher-priced curriculum.
- **Enrollment window** – the cohort spans less than two years, indicating a relatively recent intake; trends may evolve quickly.

■ Recommendations

1. **Data Enrichment** – Prioritise collection of missing demographic and payment fields (e.g., via mandatory form fields or follow-up surveys).
2. **Standardise Categorical Values** – Clean up partial entries like “Machine Learnin” and “Web Developmen” to a single canonical label to avoid fragmented reporting.
3. **Imputation Strategy** – For short-term analysis, consider imputing missing numeric values (e.g., median age) and using mode/‘Unknown’ for categorical fields, but flag imputed rows in downstream models.
4. **Focused Revenue Analysis** – Leverage the clear payment differential by course to explore pricing optimisation or targeted marketing for high-value programs.
5. **Monitor Data Quality** – Implement automated validation checks at data-ingestion time to catch missing or malformed fields before they enter the analytical pipeline.

Prepared by the autonomous AI Data Analyst.